SPEEK as a promising solid-acid catalyst for esterification, etherification, and acetylation reactions

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Scope and motivation

Commercially available polymeric acid catalysts are commonly styrene or perfluorinated resinsulfonic acid-based materials. A big challenge for these catalysts, particularly the non-halogenated ion-exchange resins, is their stability at high temperatures. [1]

For this reason, sPEEK with its high-performance thermoplastic backbone poly (ether ether) ketone (PEEK) was synthesized and investigated as an alternative catalyst. In addition to morphological studies, our research focused on the applicability of sPEEK in various reaction systems. Therefore, esterifications with sebacic acid, levulinic acid, and oleic acid were performed with different alcohols. As etherification reaction, the synthesis of dimethyl ether was investigated. [2] In addition to these high-temperature reactions between 110 °C and 170 °C, the acetylation of acetone and glycerol to solketal was tested in a temperature range between 30 °C and 50 °C.

The scope of this work is to explore the applicability of sPEEK as a solid-acid catalyst in different temperature ranges and with varying reactants. Based on the performance of these model reactions, sPEEK was tailored to obtain a chemically stable and highly active material for acid-catalyzed reactions.

Materials and methods

For catalyst manufacturing, the polymeric backbone VICTREX™ PEEK 450G™ (Victrex plc) was post-sulfonated to sPEEK. Reaction time and temperature differ between 30 to 360 minutes and between 40 °C and 60 °C, respectively. After washing and drying, the degree of sulfonation (DS) and the corresponding ion-exchange capacity (IEC) of the catalyst were analyzed by back titration with hydrochloric acid.

All esterification and acetylation reactions were carried out in a batch reactor to test the catalytic activity and stability. The etherification reactions were performed in a tubular reactor with several sampling points.

Results and discussion

The synthesized sPEEK catalysts showed IECs between 0.3 and 2.5 mmol/g, corresponding to DS between 10 % and 90 %. These values are relatively low compared to other commercial ion-exchangers such as Amberlyst™ 36 (5.4 mmol/g) and Lewatit® GF101® (4.7 mmol/g). Nonetheless, sPEEK showed high activity and, considering the degree of sulfonation, good stability in all reactions. Particularly in esterifications and acetylations, the activity of sPEEK was equal or even higher than that of the tested commercial catalysts.

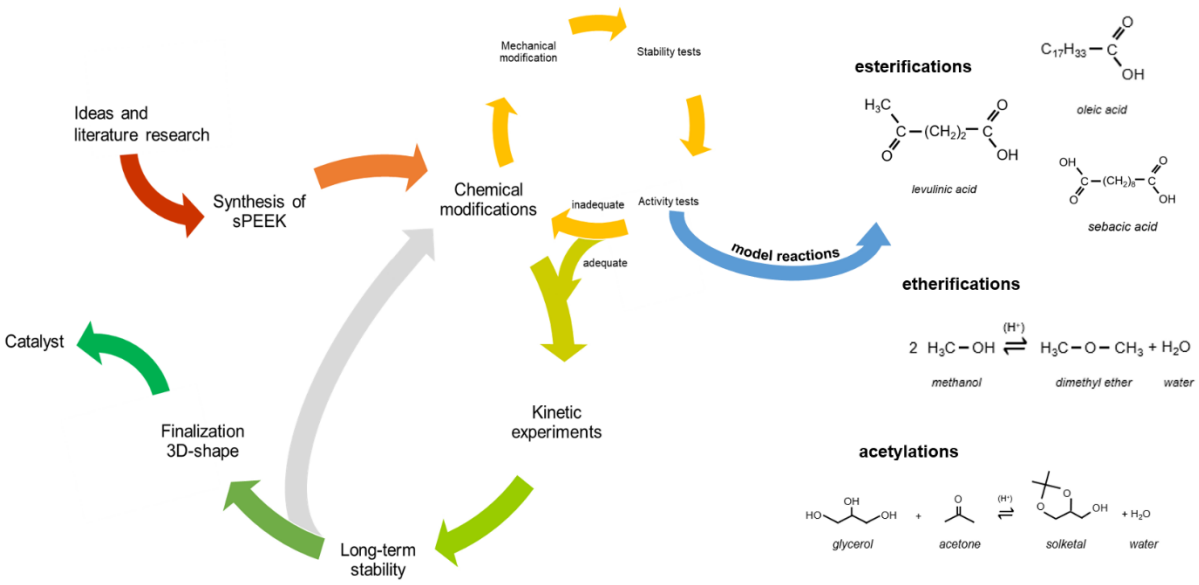

Figure 1: Overview of catalyst development and model reactions.

Conclusion

SPEEK showed promising performance in different esterification, etherification, and acetylation reactions as a non-halogenated solid-acid catalyst. Therefore, tailoring of sPEEK for heterogeneous catalysis enables new perspectives for different applications in acid catalysis.

References

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[2] A. Greve, H. Stein, T. Osterland, O. Hinrichsen, *Applied Catalysis O: Open* **2024**, *190*, 206951. DOI: https://doi.org/10.1016/j.apcato.2024.206951